

Question 1. Consider $f(x, y) = xy - 2x - y$ and $g(x, y) = 2x + y$.

(a) What is the maximum value of $f(x, y)$ subject to the constraint $g(x, y) = 20$?

$$\nabla f = \langle y-2, x-1 \rangle \quad \nabla g = \langle 2, 1 \rangle$$

$$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} y-2 = 2\lambda \\ x-1 = \lambda \end{cases} \Leftrightarrow \begin{cases} x=5 \\ y=10 \end{cases}$$

$$g(x, y) = 20 \Rightarrow 2x + y = 20 \Rightarrow \lambda = 4$$

The max is $f(5, 10) = 50 - 10 - 10 = 30$

(b) If $f(x, y)$ is the profit as a function of minutes of advertising on TV (x) and podcast (y), and $g(x, y)$ is the total cost of advertising (so xy is the revenue). What is the real-world interpretation of the Lagrange multiplier λ ? (Select multiple and argue why they must be equal.)

- ☐ The profit gained per minute of advertising on TV. ☐ The profit gained per minute of advertising on podcast.
- ☒ The profit gained per dollar of advertising on TV. ☒ The profit gained per dollar of advertising on podcast.

If they are not equal, then that means I can take the money from one and spend on the other to increase profit.

Question 2. Find the point on the graph of $y = x^2 + 4x + 6$ that is the closest to the line $y = 2x$.

First, note that they don't intersect.

Distance to the line $y - 2x = 0$ is $\frac{|y - 2x|}{\sqrt{1^2 + (-2)^2}} = \frac{|y - 2x|}{\sqrt{5}}$

give by $f(x, y) = \langle x, y \rangle \cdot \frac{\langle -2, 1 \rangle}{\|\langle -2, 1 \rangle\|} = \frac{y - 2x}{\sqrt{5}}$

$\nabla f = \langle -\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \rangle$

$\nabla g = \langle 2x+4, -1 \rangle$

$g(x, y) = 0 \Rightarrow x^2 + 4x + 6 - y = 0$

$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} -\frac{2}{\sqrt{5}} = \lambda(2x+4) \\ \frac{1}{\sqrt{5}} = \lambda(-1) \end{cases} \Leftrightarrow \begin{cases} x = -1 \\ y = 3 \end{cases}$

The answer is $\boxed{(-1, 3)}$.

The constraint is $g(x, y) = x^2 + 4x + 6 - y = 0$.

Question 3. Consider the tilted ellipse $5x^2 - 6xy + 5y^2 = 4$. Find the points on the ellipse that are closest to and farthest from the origin.

Distance to origin is $f(x, y) =$

We can extremize the squared distance to origin $f(x, y) = x^2 + y^2$.

The constraint is $g(x, y) = 5x^2 - 6xy + 5y^2 = 4$.

$$\nabla f = \langle 2x, 2y \rangle$$

$$\nabla g = \langle 10x - 6y, -6x + 10y \rangle$$

$$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} 2x = \lambda(10x - 6y) \\ 2y = \lambda(-6x + 10y) \end{cases} \Rightarrow \begin{cases} 2x = \lambda(10x - 6y) \\ 2y = \lambda(-6x + 10y) \end{cases}$$

Two cases:

① $\lambda = \frac{1}{2}, x = y = \pm 1$

② $\lambda = \frac{1}{5}, x = \pm \frac{1}{2}, y = \mp \frac{1}{2}$

We calculate $f(\pm 1, \pm 1) = 2 \Rightarrow (1, 1) \text{ and } (-1, -1) \text{ are furthest}$

$f(\pm \frac{1}{2}, \mp \frac{1}{2}) = \frac{1}{2} \Rightarrow (\frac{1}{2}, -\frac{1}{2}) \text{ and } (-\frac{1}{2}, \frac{1}{2}) \text{ are closest}$

Question 4 (Second derivative test). Find the critical points of $f(x, y) = x^2 + 3xy + y^2$ and determine whether they are local maxima, minima, or saddle points.

$$\nabla f = \langle 2x + 3y, 3x + 2y \rangle = 0 \Leftrightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

The only critical point is $(0, 0)$.

$$\partial_{xx} f = 2$$

$$\partial_{xy} f = \partial_{yx} f = 3$$

$$\partial_{yy} f = 2$$

$$\begin{vmatrix} \partial_{xx} f & \partial_{xy} f \\ \partial_{yx} f & \partial_{yy} f \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5 < 0 \Rightarrow (0, 0) \text{ is a saddle point}$$

Question 5. The following boxed arguments are incorrect or incomplete. Identify the error in each argument and provide a correct solution.

Question: Determine whether $\sum_{n=2}^{\infty} \frac{1}{(\ln(n)+2)^2}$ converges.

Flawed Argument: Yes, because as n approaches infinity, $\ln(n) + 2$ approaches infinity, so $\frac{1}{(\ln(n)+2)^2}$ approach zero which is less than one. *This is not a test!*

Compare No, diverges by comparing with $\sum_{n=2}^{\infty} \frac{1}{n}$.

$\lim_{n \rightarrow \infty} \frac{(\ln(n)+2)^2}{n} = \left(\lim_{n \rightarrow \infty} \frac{\ln(n)+2}{\sqrt{n}}\right)^2 \rightarrow 0$ because $\ln$ is positive power. Since $\frac{1}{(\ln(n)+2)^2} \gg \frac{1}{n}$ and $\sum \frac{1}{n}$ diverges, the origin series diverges.

Question: Find the interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{(x-2)^n}{5n+3}$.

Flawed argument: Apply ratio test, the series converges when

$$\lim_{n \rightarrow \infty} \left| \frac{(x-2)^{n+1}}{5(n+1)+3} \cdot \frac{5n+3}{(x-2)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{5n+3}{5n+8} (x-2) \right| = |x-2| < 1.$$

Thus, $-1 < x-2 < 1$ which is equivalent to $1 < x < 3$, so the interval of convergence is $(1, 3)$.

Just need to check endpoints!

For $x=1$, $\sum_{n=0}^{\infty} \frac{(x-2)^n}{5n+3} = \sum_{n=0}^{\infty} \frac{(-1)^n}{5n+3}$ converges by AST ($\frac{1}{5n+3}$ is decreasing and tends to zero (and nonnegative))

For $x=3$, $\sum_{n=0}^{\infty} \frac{(x-2)^n}{5n+3}$ diverges by comparing with $\sum \frac{1}{n}$ (...). Thus the interval of convergence is $[1, 3)$.

Question: Calculate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$ or show that it does not exist.

Flawed argument: Along the x -axis, the limit is $\lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2+0} = 0$ and along the y -axis, the limit is $\lim_{y \rightarrow 0} \frac{0 \cdot y}{0+y^2} = 0$, so the limit is 0.

Existence requires convergence for all paths, not just two. Nonexistence only requires different limit for two paths.

Does not exist because along $y=x$, we have $\lim_{x \rightarrow 0} \frac{x \cdot x}{x^2+x^2} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \frac{1}{2}$ which is a different limit from other paths (for example, along the x -axis).

Question: Find the tangent line to the curve $y = y^2 - x^3 + 1$ at the point $(1, 0)$.

Flawed argument: We compute the gradient of $f(x, y) = y^2 - x^3 + 1$ which at the point $(1, 0)$ is

$$\nabla f(x, y)|_{(1,0)} = \langle -3x^2, 2y \rangle|_{(1,0)} = \langle -3, 0 \rangle.$$

Note that we need to interpret the graph as $f(x,y)=c$ for some constant c . The original graph is $y^2 - x^3 + 1 = y$ and y is not a constant.

The tangent line is the line orthogonal to the gradient and passing through the point $(1, 0)$, so the equation of the line is $\langle x-1, y-0 \rangle \cdot \langle -3, 0 \rangle = 0$ which is simply $x = 1$.

Instead, we use $f(x,y) = y^2 - x^3 + 1 - y$ and compute $\nabla f(1,0) = \langle -3x^2, 2y-1 \rangle|_{(1,0)} = \langle -3, -1 \rangle$

The tangent line is $\langle x-1, y-0 \rangle \cdot \langle -3, -1 \rangle = 0$ which is $-3(x-1) - y = 0$ or $y = -3(x-1)$.

Question: Let $f(x, y) = (x-1)y$ and $x(t) = e^t$ and $y(t) = \cos(y)$. Find the first two nonzero terms of the Maclaurin series of $f(x(t), y(t))$.

Flaws argument: We write $x(t) = 1 + t$ and $y(t) = 1 - \frac{t^2}{2}$, then we get

$$f(x(t), y(t)) = (1+t-1) \left(1 - \frac{t^2}{2}\right) = t - \frac{t^3}{2}.$$

cannot write equal! We have to write $x(t) = 1 + t + \dots$ and $y(t) = 1 - \frac{t^2}{2}$. We then realize that $f(x(t), y(t)) = (t + \dots) \left(1 - \frac{t^2}{2} + \dots\right) = t + \dots$ because we don't know higher powers due to the possibility that there is t^2 in the first factor.

$x(t) = 1 + t + \frac{t^2}{2} + \dots$ $y(t) = 1 - \frac{t^2}{2} + \dots$ (this turns out to be all we need, but you might want to write more terms)

$$f(x(t), y(t)) = \left(t + \frac{t^2}{2} + \dots\right) \left(1 - \frac{t^2}{2} + \dots\right) = t + \frac{t^2}{2} + \dots$$

It would be wrong to write $-\frac{t^3}{2}$ because there might be other t^3 terms in the ellipses that we have not written down.